

Assignment-2

Applying micro services and Event-Driven Architecture in E-commerce

Scenario: Imagine developing an e-commerce platform with the

Following architecture.

• Microservice Architecture:

• Services - User Management, Product catalog, shopping Cart, Order Processing, Payment and Notification

• Benefits: Each service can be developed, deployed and scaled independently, enhancing flexibility and resilience.

• Events - Driven Architecture

• Mechanism: Services communication through events [eg: "Order Placed", "Payment Processed"]

• Benefits: Promote loose coupling and allow services to react to events asynchronously, improving scalability & responsiveness.

Applying Solid principles

• Single responsibility: Each microservice handles a specific business function.

• Open / closed principle: Services can be extended with new feature without modifying existing code.

• Liskov Substitution Principle: Interchangeable service implementations without affecting system's behaviour.

Observing Dry and KISS Principle:

*Dry [Don't Repeat Yourself]

- Utilize shared libraries for common functionalities like logging and authentication.
- Centralize configuration management to avoid redundancy.

*KISS [keep it simple, stupid]

- Design services with clear and concise responsibilities.
- Avoid over-engineering - implement feature as needed.